

SENSORS That Add Strength To AVIATION And AEROSPACE

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Aviation is witnessing fast growth in global markets. Whether it is civil or military aviation, modern aircraft are equipped with electronics systems

Accurate feedback systems and ease of control of aircraft systems are enabled by electronic sensors. Sensors help in measuring various parameters like monitoring, control and navigation. Thus, avionics or electronic systems play an important role in modern aviation.

Avionic systems include systems for communication, navigation and display, searchlights and complex tactical systems for airborne early warning signals. For example, Lockheed SR-71 Blackbird—world's fastest and highest-flying operational manned, air-breathing jet aircraft, which was used by the US Air Force—is equipped with electronic systems including signal intelligence sensors, side-looking airborne radars and camera sensors. If it detects a missile launched by the enemy, it simply accelerates and outflies the missile.

All sensors installed in aircraft are for flight instruments. These include tachometers, engine temperature gauges, fuel- and



Fig. 1: SR-71 Blackbird, world's fastest and highest-flying operational manned, air-breathing jet aircraft (Credit: <https://en.wikipedia.org>)

oil-quantity gauges, pressure gauges, altimeters, airspeed-measurement meters, vertical speed indicators and others. Then, there are sensors for ground testing, flight testing, vibration, environment, angle of attack and static. There are also doppler radars, lightning-detection radars, terrain radars, anti-collision warning systems and stall warning systems. Many of these instruments and control sensors supply additional signals to cockpit indicators, informing the pilots to take proper action and precaution, and prevent any kind of disaster or accident.

Aircraft computer systems receive data from various sensors, including air temperature probes, angle-of-attack probes and pitot-static pressure systems. These computers process inputs from these sensors, apply compensating factors and transmit information to the displays in the cockpit. The pilots continuously monitor the status of the engine and environment from the cockpits.

There are thousands of sensors for different commercial aircraft. It is too broad a category, not to mention the hundreds of models and makes of planes covering different generations of aircraft. We have



Fig. 2: Cockpit of Boeing 757-300 (Credit: <https://commons.wikimedia.org>)